Quectel EG95&EG91 Series Module

Introduction

June, 2020

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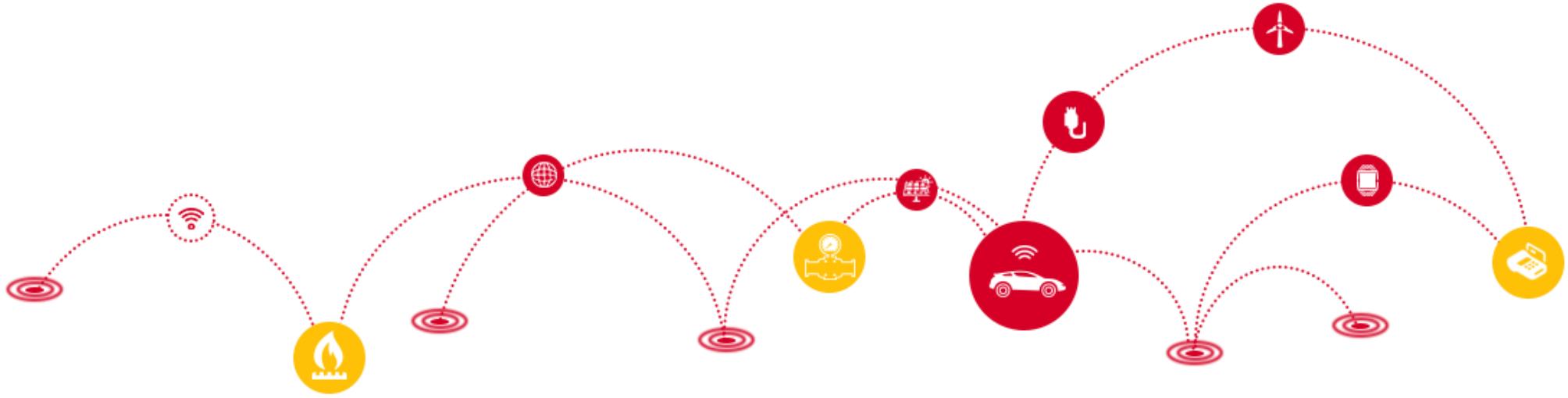
Roadmap

Specifications & Highlights

Development Timeline

Technical Details

Applications



EG95&EG91 Series Roadmap

Product Lifetime > 2025

Cat 4

EG95-E
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- EMEA

EG95-NA
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- North America

X Series

EG95-NAX
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- North America

EG95-EX
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- EMEA

EG95-NAXD (Data-only)
Nov. 2019 NEW
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- North America

EG95-AUX
Q3. 2020 NEW
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- Latin America/ Australia/ New Zealand

Cat 1

EG91-E
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- EMEA

EG91-NS/ -NA
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- North America

EG91-NAX
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- North America

EG91-EX
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- EMEA

EG91-NAXD (Data-only)
Nov. 2019 NEW
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- North America

EG91-AUX
Q3. 2020 NEW
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- Latin America/ Australia/ New Zealand

EG91-VX
- Compatible with BG96/ BG95 Series/ BC95-G/ UG9x/ M95
- Multi-mode LTE
- North America

MM.YYYY
Qx.YYYY
Estimated Engineering Sample Dates

2016 | 2017 | 2018 | 2019/2020

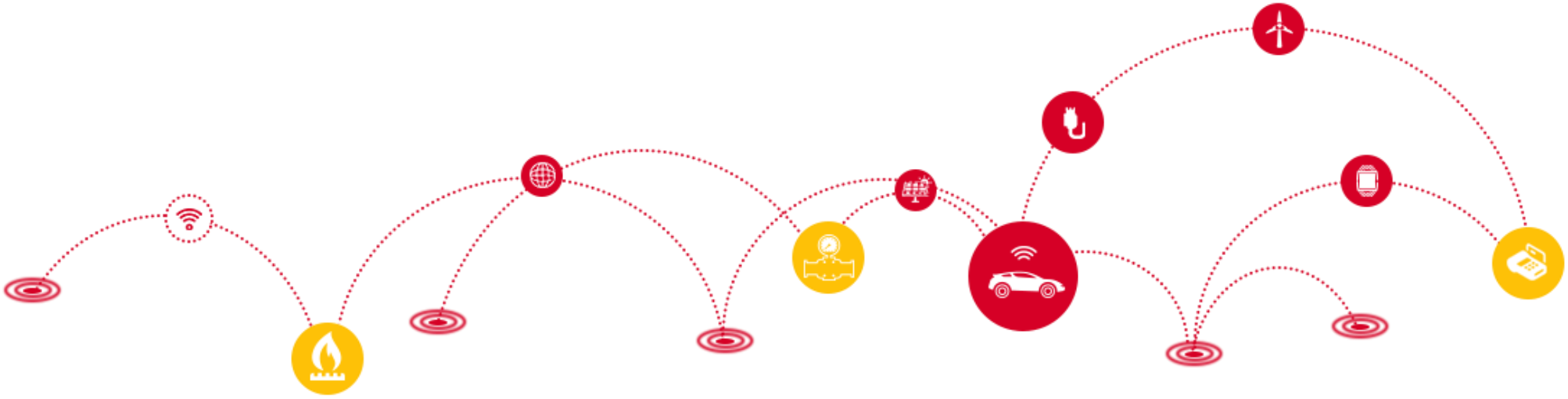
Roadmap

Highlights & Specifications

Development Timeline

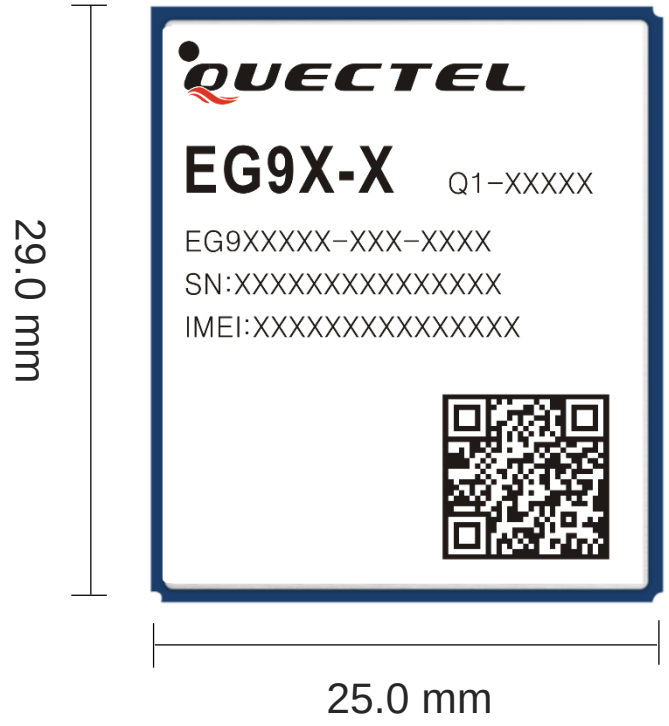
Technical Details

Applications

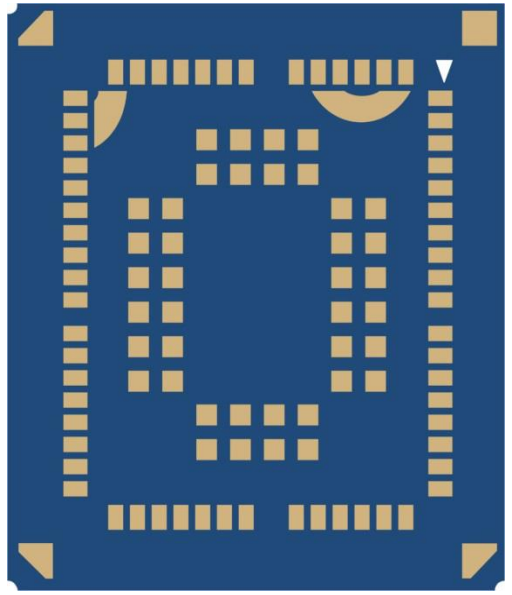


EG95/EG91 Series Mechanical Dimensions

Multi-mode LTE Module (MDM9x07)







Length: 29.0 mm (±0.15 mm)
Width: 25.0 mm (±0.15 mm)
Height: 2.3 mm (±0.2 mm)
Weight: Approx. 3.8 g

EG95/EG91 Series Highlights

Highlight	Description
Hardware Interfaces	USB 2.0/ PCM/ (U)SIM/ I2C/ UART/ Antennas①/ NETNIGHT/ PWRKEY/ RESET_N/ SPI/ USB_BOOT
Form Factor	LGA
Multi-band	LTE, UMTS/HSPA+ and GSM/GPRS/EDGE coverage
GNSS② (Optional)	GPS, GLONASS, BeiDou/Compass, Galileo, QZSS
DFOTA	Delta Firmware Upgrade Over-The-Air
QuecOpen®③	100 MB Flash and 100 MB RAM
QuecLocator®④	Supported
(U)SIM Card Detection	Supported
Compatibility	Compatible with Quectel M95/UG95/UG96/BC95-G/BG95 series/BG96 modules
Abundant Protocols	TCP /UDP/ FTP /FTPS/ HTTP/ HTTPS/ NTP/ PING/ QMI/ NITZ/ MMS/ SMTP/ SSL/ MQTT/ FILE/ PPP/ SMTPS/ CMUX
Certification	Comprehensive certifications

① GNSS antenna is not supported by EG95-E and EG91-E modules.
Rx-diversity antenna is not supported by EG91-AUX module.
② GNSS function is not supported by EG95-E and EG91-E modules.
③ QuecOpen® is not supported by X Series modules.
④ means QuecLocator® function is not supported by EG91-E and EG95-E modules.

EG95 Series Specifications



29.0 mm × 25.0 mm × 2.3 mm
150 Mbps DL/ 50 Mbps UL

Multi-mode LTE Cat 4 Module

Variant		EG95-E	EG95-EX	EG95-NA	EG95-NAX	EG95-NAXD (Data-only) 2020, New	EG95-AUX 2020, New
LTE	LTE-FDD	B1/ B3/ B7/ B8/ B20/ B28A	B1/ B3/ B7/ B8/ B20/ B28	B2/ B4/ B5/ B12/ B13	B2/ B4/ B5/ B12/ B13/ B25/ B26	B2/ B4/ B5/ B12/ B13/ B25/ B26	B1/ B2/ B3/ B4/ B5/ B7/ B8/ B28/ B66
	LTE-TDD	\	\	\	\	\	\
UMTS	WCDMA	B1/ B8	B1/ B8	B2/ B4/ B5	B2/ B4/ B5	B2/ B4/ B5	B1/ B2/ B5/ B8
	TD-SCDMA	\	\	\	\	\	\
EVDO/CDMA		\	\	\	\	\	\
GSM/EDGE		B3/ B8	B3/B8	\	\	\	Quad-band
GNSS		\	Optional	Optional	Optional	Optional	Optional
Wi-Fi/BT Interface		\	\	\	\	\	\
Ethernet Interface		\	\	\	\	\	\
Region		EMEA	EMEA	North America	North America	North America	Latin America/ Australia/ New Zealand
Certification		**Regulatory:** GCF/ CE/ RCM/ FAC **Others:** WHQL	**Regulatory:** GCF/ CE/ RCM **Others:** WHQL	**Carrier:** Verizon/ AT&T/ T-Mobile/ U.S. Cellular/ Rogers/ Telus **Regulatory:** GCF/ FCC/ PTCRB/ IC **Others:** WHQL	**Carrier:** Sprint/ Verizon*/ AT&T*/ T-Mobile*/ U.S. Cellular* **Regulatory:** GCF/ FCC/ PTCRB/ IC **Others:** WHQL	**Carrier:** Verizon*/ AT&T*/ T-Mobile*/ Sprint* **Regulatory:** GCF/ FCC*/ PTCRB* **Others:** WHQL	**Regulatory:** Anatel*/ RCM* **Others:** WHQL

"" means under development.*

EG91 Series Specifications



29.0 mm × 25.0 mm × 2.3 mm
10 Mbps DL/ 5 Mbps UL

Multi-mode LTE Cat 1 Module

Variant		EG91-E	EG91-EX	EG91-NA	EG91-NAX	EG91-NAXD (Data-only) 2020, New	EG91-NS	EG91-VX	EG91-AUX 2020, New
LTE	LTE-FDD	B1/ B3/ B7/ B8/ B20/ B28A	B1/ B3/ B7/ B8/ B20/ B28	B2/ B4/ B5/ B12/ B13	B2/ B4/ B5/ B12/ B13/ B25/ B26	B2/ B4/ B5/ B12/ B13/ B25/ B26	B2/ B4/ B5/ B12/ B13/ B25/ B26	B4/ B13	B1/ B2/ B3/ B4/ B5/ B7/ B8/ B28/ B66
	LTE-TDD	\	\	\	\	\	\	\	\
UMTS	WCDMA	B1/ B8	B1/ B8	B2/ B4/ B5	B2/ B4/ B5	B2/ B4/ B5	B2/ B4/ B5	\	B1/ B2/ B5/ B8
	TD-SCDMA	\	\	\	\	\	\	\	\
EVDO/CDMA		\	\	\	\	\	\	\	\
GSM/EDGE		B3/ B8	B3/B8	\	\	\	\	\	Quad-band
GNSS		\	Optional	Optional	Optional	Optional	Optional	Optional	Optional
Wi-Fi/BT Interface		\	\	\	\	\	\	\	\
Ethernet Interface		\	\	\	\	\	\	\	\
Region		EMEA	EMEA	North America	North America	North America	North America	North America	Latin America/ Australia/ New Zealand
Certification		**Carrier:** Deutsche Telekom **Regulatory:** GCF/ CE/ RCM/ Anatel/ FAC **Others:** WHQL	**Regulatory:** GCF/ CE/ RCM/ NCC* **Others:** WHQL	**Carrier:** Verizon/ AT&T/ T-Mobile/ U.S. Cellular/ Rogers/ Telus **Regulatory:** GCF/ FCC/ PTCRB/ IC **Others:** WHQL	**Carrier:** AT&T/ Sprint/ Verizon*/ T-Mobile*/ U.S. Cellular* **Regulatory:** GCF/ FCC/ PTCRB/ IC **Others:** WHQL	**Carrier:** Verizon*/ AT&T*/ T-Mobile*/ Sprint* **Regulatory:** GCF/ FCC*/ PTCRB* **Others:** WHQL	**Carrier:** Verizon/ AT&T/ Sprint/ T-Mobile* **Regulatory:** GCF/ FCC/ PTCRB/ IC **Others:** WHQL	**Carrier:** Verizon* **Regulatory:** GCF/ FCC **Others:** WHQL	**Regulatory:** Anatel*/ RCM* **Others:** WHQL

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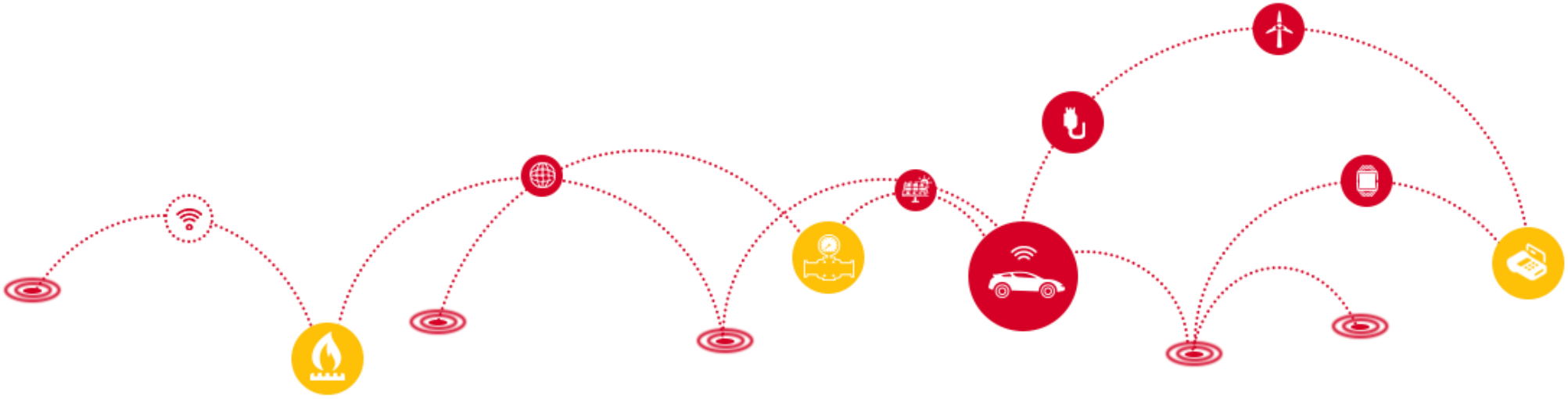
Roadmap

Specifications & Highlights

Development Timeline

Technical Details

Applications



EG95-E Timeline

2019								2020		
May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.

Project Stage

EG95-E MP

Regulatory Certifications

GCF/ CE/ RCM/ FAC

Others

WHQL

EG91-E Timeline

2019								2020		
May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.

Project Stage

EG91-E MP

Carrier Certification

Deutsche Telekom

Regulatory Certifications

GCF/ CE/ RCM/ Anatel/ FAC

Others

WHQL

EG91-NA/EG95-NA Timeline

2019											
Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.

Project Stage

EG91-NA/EG95-NA MP

Carrier Certifications

Verizon/ AT&T/ T-Mobile/ U.S. Cellular/ Rogers/ Telus

Regulatory Certifications

GCF/ FCC/ PTCRB/ IC

Others

WHQL

EG95-NAX Timeline

2019				2020							
Sep.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.

Project Schedule



ES: Engineering samples ready. Basic functions are available for customers' simple demo purpose.
CS: Commercial samples ready. Stable hardware design and quite stable software design. New software features can be added upon request.
MP: Hardware and software ready for mass production. For certification status, please refer to the "certification schedule".

Carrier Certification Schedule



Regulatory Certifications

GCF/ FCC/ PTCRB/ IC Completed

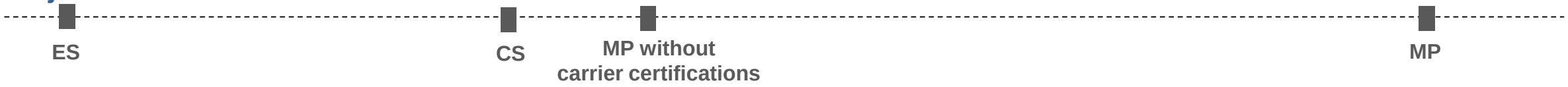
Others

WHQL Completed

EG95-NAXD Timeline

2019		2020									
Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.

Project Schedule



ES: Engineering samples ready. Basic functions are available for customers' simple demo purpose.
CS: Commercial samples ready. Stable hardware design and quite stable software design. New software features can be added upon request.
MP: Hardware and software ready for mass production. For certification status, please refer to the "certification schedule".

Carrier Certification Schedule



Regulatory Certification Schedule



Others

WHQL Completed

EG91-NAX Timeline

2019				2020							
Sep.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.

Project Schedule



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CS: Commercial samples ready. Stable hardware design and quite stable software design. New software features can be added upon request.
MP: Hardware and software ready for mass production. For certification status, please refer to the "certification schedule".

Carrier Certification Schedule

AT&T/ Sprint **Completed**

Verizon



T-Mobile/ U.S. Cellular **TBD**

Regulatory Certifications

GCF/ FCC/ PTCRB/ IC **Completed**

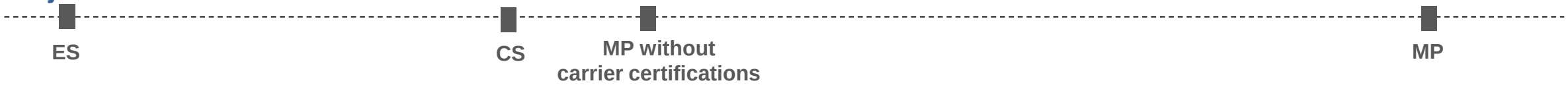
Others

WHQL **Completed**

EG91-NAXD Timeline

2019		2020									
Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.

Project Schedule



ES: Engineering samples ready. Basic functions are available for customers' simple demo purpose.
CS: Commercial samples ready. Stable hardware design and quite stable software design. New software features can be added upon request.
MP: Hardware and software ready for mass production. For certification status, please refer to the "certification schedule".

Carrier Certification Schedule



Regulatory Certification Schedule

GCF Completed



Others

WHQL Completed

EG95-EX Timeline

2019								2020			
May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.

Project Stage

EG95-EX MP

Regulatory Certifications

GCF/ CE/ RCM

Others

WHQL

EG91-EX Timeline

2019			2020								
Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.

Project Stage

EG91-EX MP

Regulatory Certification Schedule

GCF/ CE/ RCM **Completed**

NCC **Start** **Complete (Planned)**

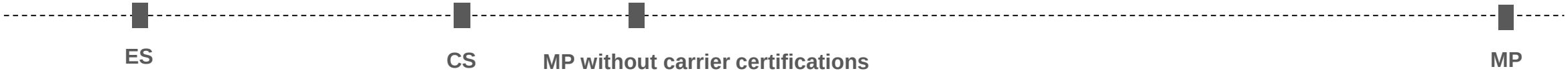
Others

WHQL **Completed**

EG91-NS Timeline

2019												2020
Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.	Jan.

Project Schedule



ES: Engineering samples ready. Basic functions are available for customers' simple demo purpose.
CS: Commercial samples ready. Stable hardware design and quite stable software design. New software features can be added upon request.
MP: Hardware and software ready for mass production. For certification status, please refer to the "certification schedule".

Carrier Certification Schedule

Verizon/ AT&T/ Sprint **Completed**

T-Mobile **TBD**

Regulatory Certifications

GCF/ FCC/ PTCRB/ IC **Completed**

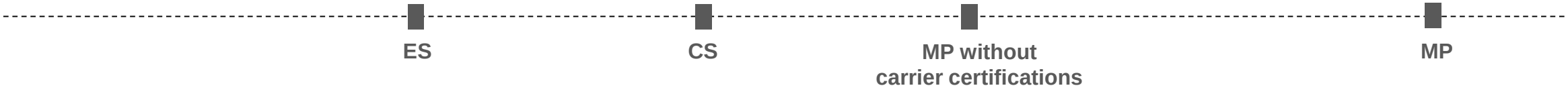
Others

WHQL **Completed**

EG91-VX Timeline

2019					2020						
Aug.	Sep.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.

Project Schedule



ES: Engineering samples ready. Basic functions are available for customers' simple demo purpose.
CS: Commercial samples ready. Stable hardware design and quite stable software design. New software features can be added upon request.
MP: Hardware and software ready for mass production. For certification status, please refer to the "certification schedule".

Carrier Certification Schedule



Regulatory Certification

GCF/ FCC **Completed**

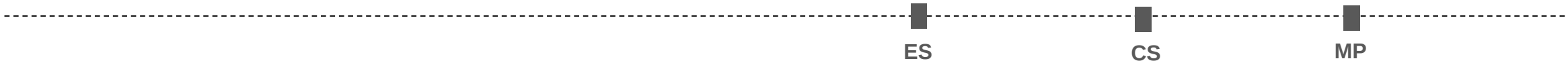
Others

WHQL **Completed**

EG95-AUX/EG91-AUX Timeline

2020											
Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.

Project Schedule



ES: Engineering samples ready. Basic functions are available for customers' simple demo purpose.
CS: Commercial samples ready. Stable hardware design and quite stable software design. New software features can be added upon request.
MP: Hardware and software ready for mass production. For certification status, please refer to the "certification schedule".

Regulatory Certification Schedule



Others

WHQL **Completed**

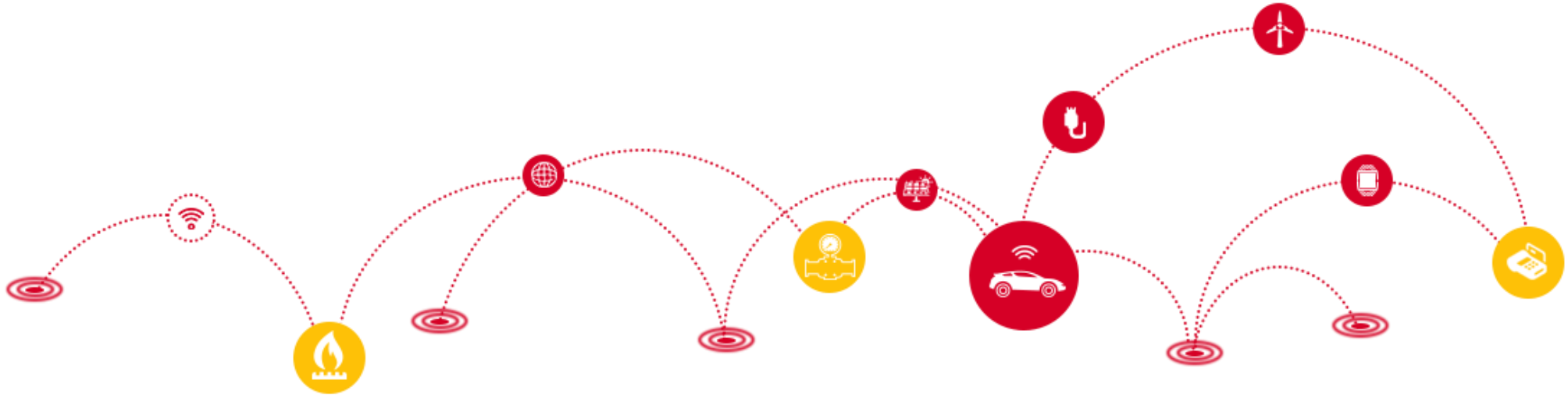
Roadmap

Specifications & Highlights

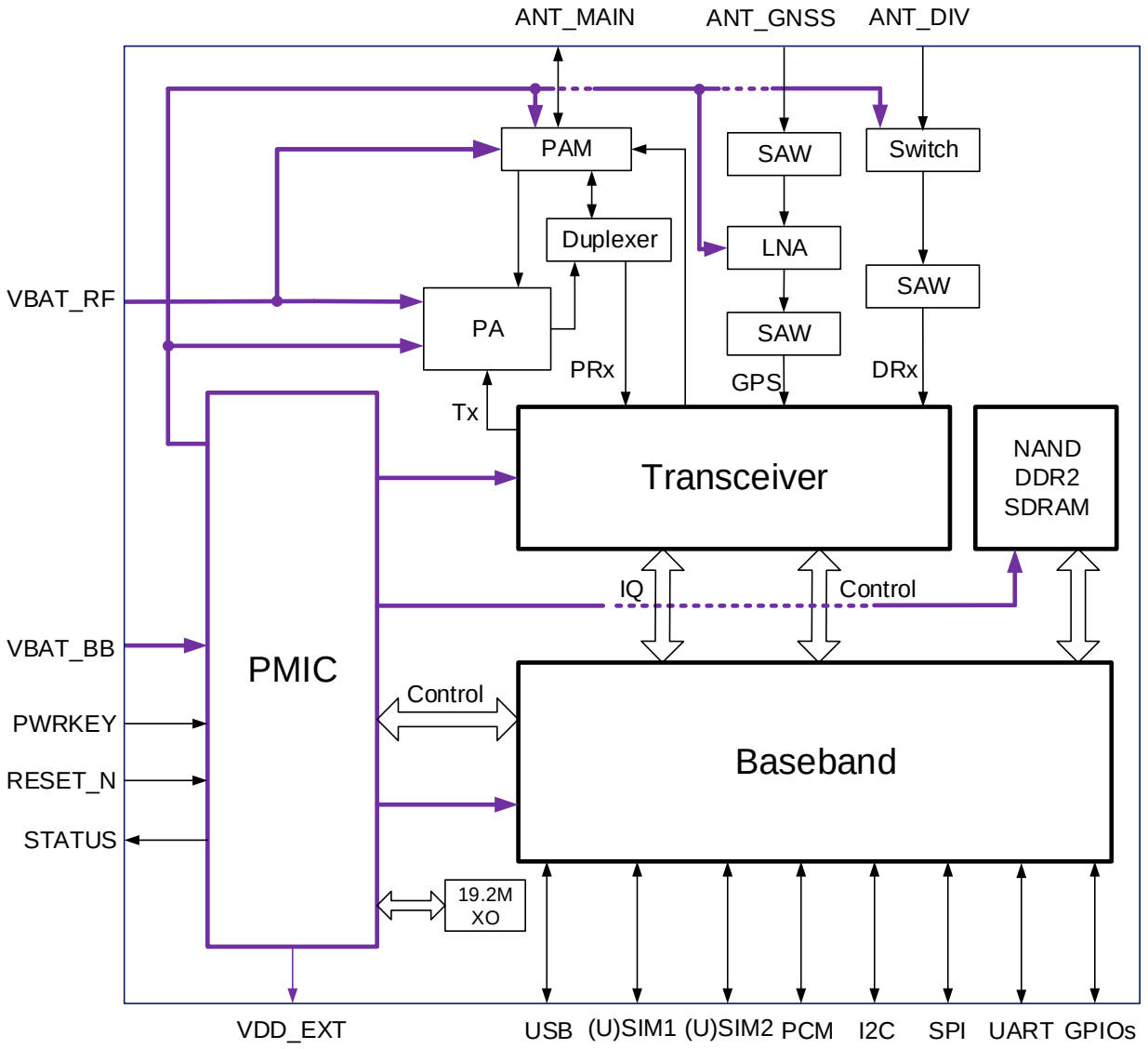
Development Timeline

Technical Details

Applications



Hardware Architecture



NOTES: GNSS antenna is not supported by EG95-E and EG91-E modules.
Rx-diversity antenna is not supported by EG91-AUX module.

Software Advantages

USB Serial Driver

- Windows 7/8/8.1/10
- Linux 2.6–5.4
- Android 4.x–9.x

Other Drivers

- GNSS① Driver: Android 4.x–9.x
  RIL Driver: Android 4.x–10.x
- NDIS Driver:
  Windows 7/8/8.1/10
- MBIM Driver①:
  Windows 8/8.1/10, Linux 3.18–5.4
- GobiNet Driver:
  Linux 2.6–5.4
- QMI_WWAN Driver:
  Linux 3.4–5.4

Quality Guarantee

- Reliable network protocols
- Steady flash protection mechanism
- Superior audio algorithms
- High sensitivity

Flexible Applications

- (U)SIM card detection
- DTMF

Special Features

- QuecOpen®②
- QuecLocator®③
- LwM2M
- eMBMs (Optional)
- Secure boot*

Enhanced AT Commands

- Standard V.25ter AT commands
- 3GPP TS 27.007 (GSM 07.07)
- 3GPP TS 27.005 (GSM 07.05 SMS)
- TCP/IP stack AT commands
- STK (SIM Application Toolkit)

"" means under development.*
① means GNSS driver is not supported by EG91-E and EG95-E modules.
MBIM driver is not supported by X Series modules.
② means QuecOpen® is not supported by X Series modules.
③ means QuecLocator® function is not supported by EG91-E and EG95-E modules.

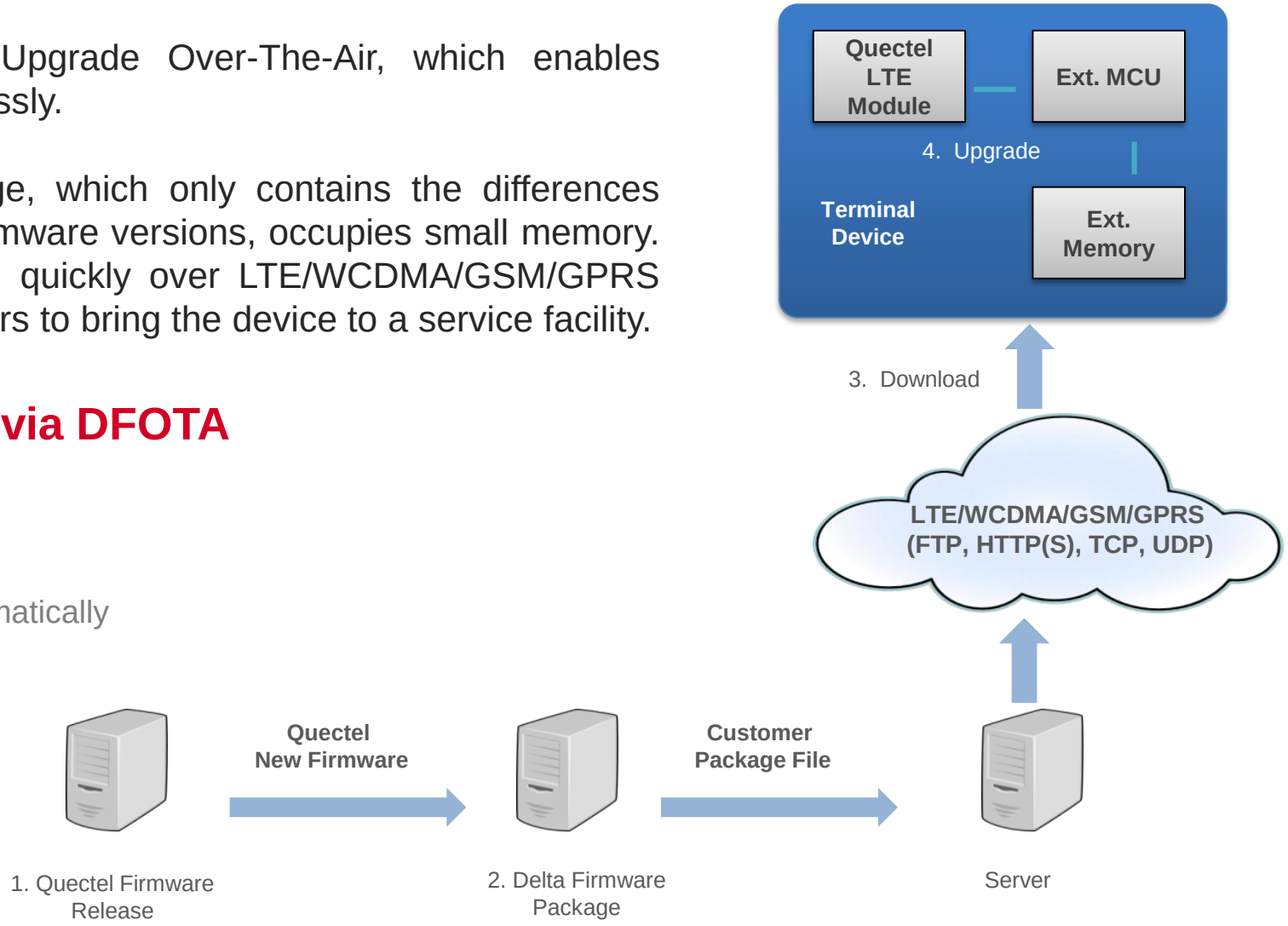
DFOTA

DFOTA refers to Delta Firmware Upgrade Over-The-Air, which enables customers to upgrade firmware wirelessly.

In DFOTA, a delta firmware package, which only contains the differences between the source and the target firmware versions, occupies small memory. Thus the firmware can be upgraded quickly over LTE/WCDMA/GSM/GPRS networks, eliminating the need for users to bring the device to a service facility.

Firmware Upgrade Process via DFOTA

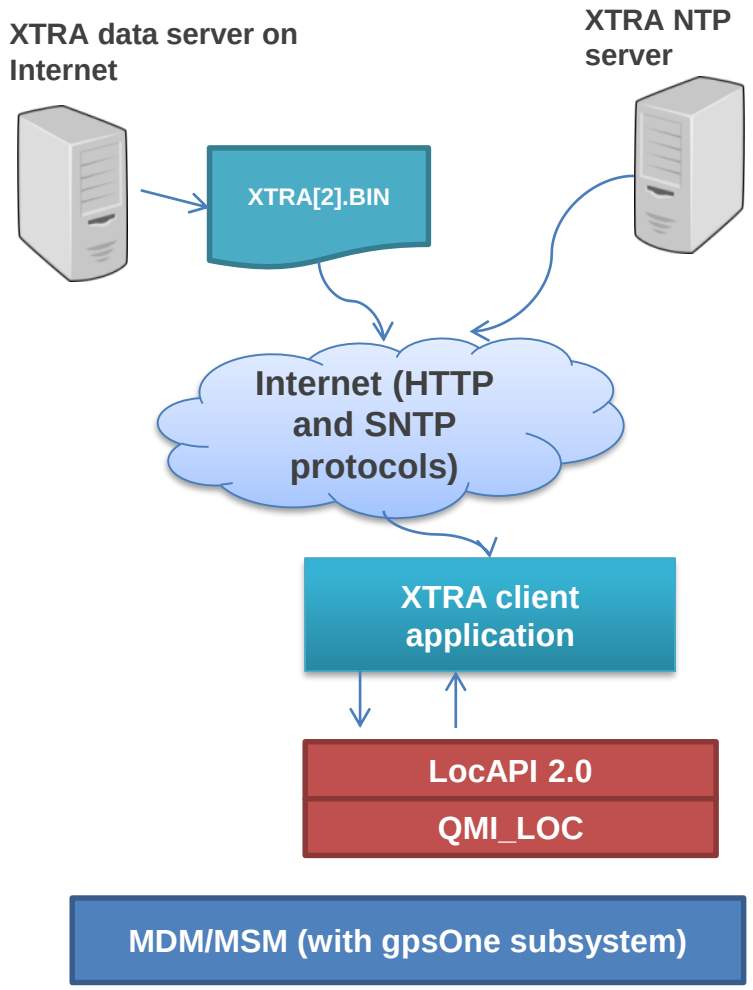
- Get Delta Firmware Package
- Put Delta Package on Server
- Execute **AT+QFOTADL** Command
- Download and Upgrade Firmware Automatically



XTRA Assistance Technology

Qualcomm's new gpsOne XTRA Assistance technology provides enhanced operation by enabling a user to download a small assistance data file through a brief Internet access session.

This technology delivers more accurate positioning with greater sensitivity than otherwise possible with standalone GPS receivers, especially in challenging areas such as indoors and in dense urban canyons.



GNSS Performance of EG95 & EG91 Series

Module			EG91 Series	EG95 Series
GNSS			GPS, GLONASS, BeiDou/Compass, Galileo, QZSS	GPS, GLONASS, BeiDou/Compass, Galileo, QZSS
Sensitivity (GNSS)	Cold start		-146 dBm	-146 dBm
	Reacquisition		-157 dBm	-157 dBm
	Tracking		-157 dBm	-157 dBm
TTFF (GNSS)	Cold start @open sky	Autonomous	35 s	35 s
		XTRA enabled	12 s	12 s
	Warm start @open sky	Autonomous	26 s	26 s
		XTRA enabled	3.7 s	3.8 s
	Hot start @open sky	Autonomous	2 s	2 s
		XTRA enabled	3.4 s	3.2 s
SVs	Total number of SVs available		55	55
Accuracy (GNSS)	CEP-50		< 2.5 m	< 2.5 m

NOTE: GNSS function is not supported by EG95-E and EG91-E modules.

QuecOpen®

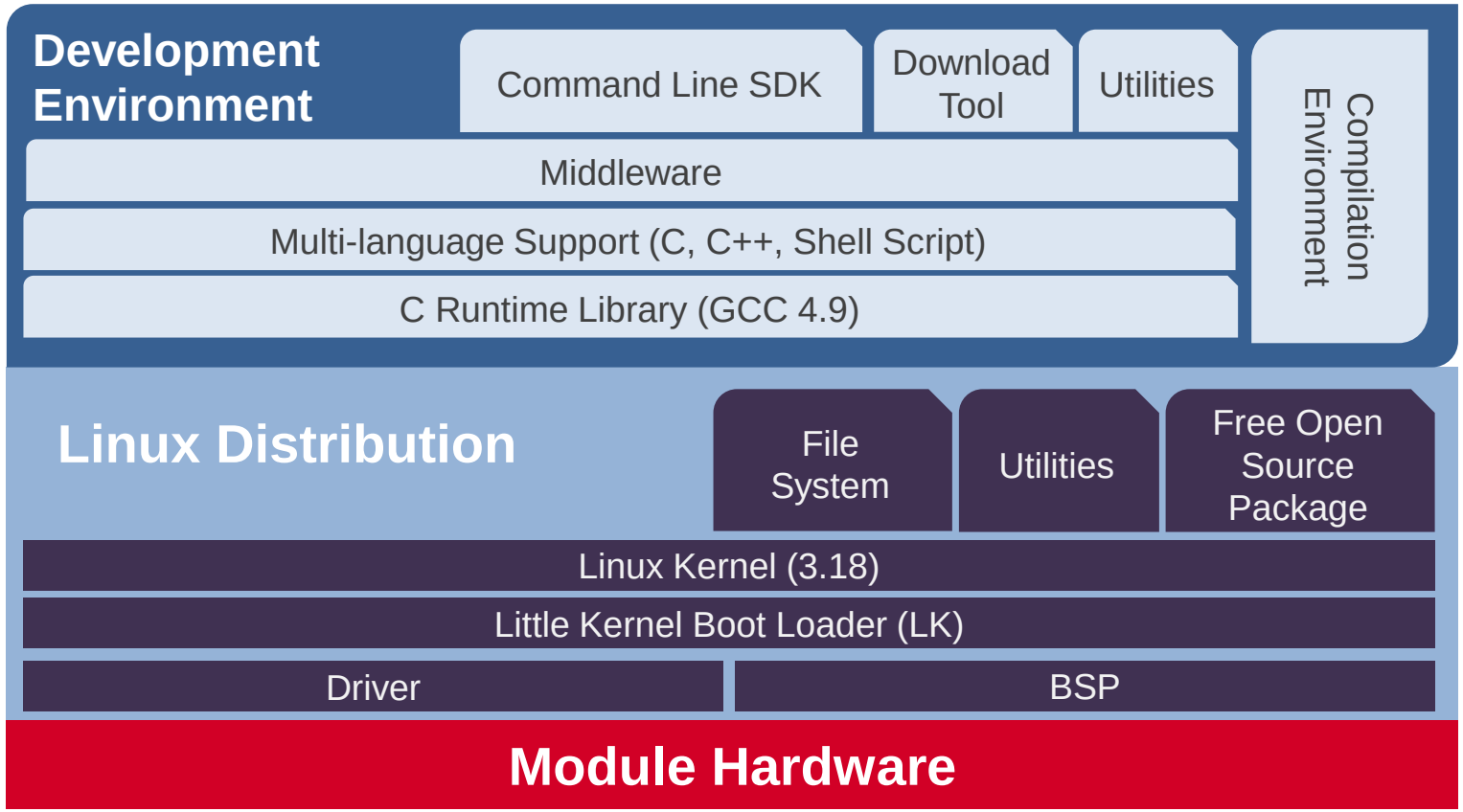


Figure: Framework of QuecOpen®

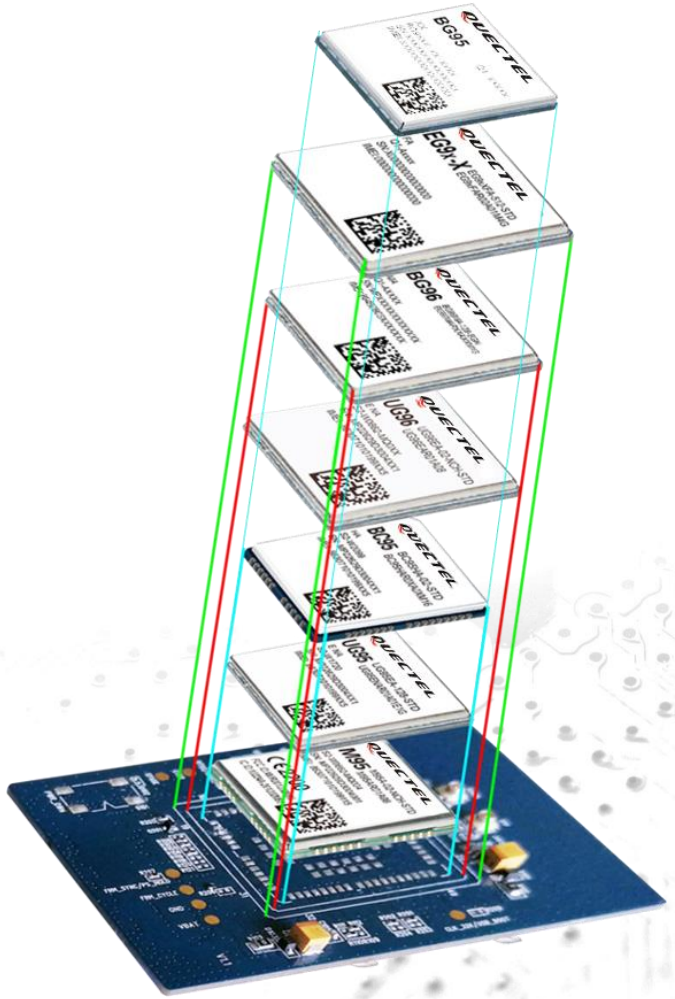
QuecOpen® is an open source embedded development platform based on Linux system, which is intended to simplify the design and development process for IoT applications.

NOTE: QuecOpen® function is optional is not supported by X Series modules.

Design Compatibility

EG9x series module is compatible with the footprints of the following Quectel modules:

- GSM/GPRS module M95
- UMTS/HSPA modules UG95 & UG96
- LTE Cat NB2 module BC95-G
- LTE Cat M1/Cat NB2/EGPRS module BG95 Series
- LTE Cat M1/Cat NB1/EGPRS module BG96



NOTE: The compatibility diagram shown above is for illustration purpose only. The actual appearance of the modules may be different.

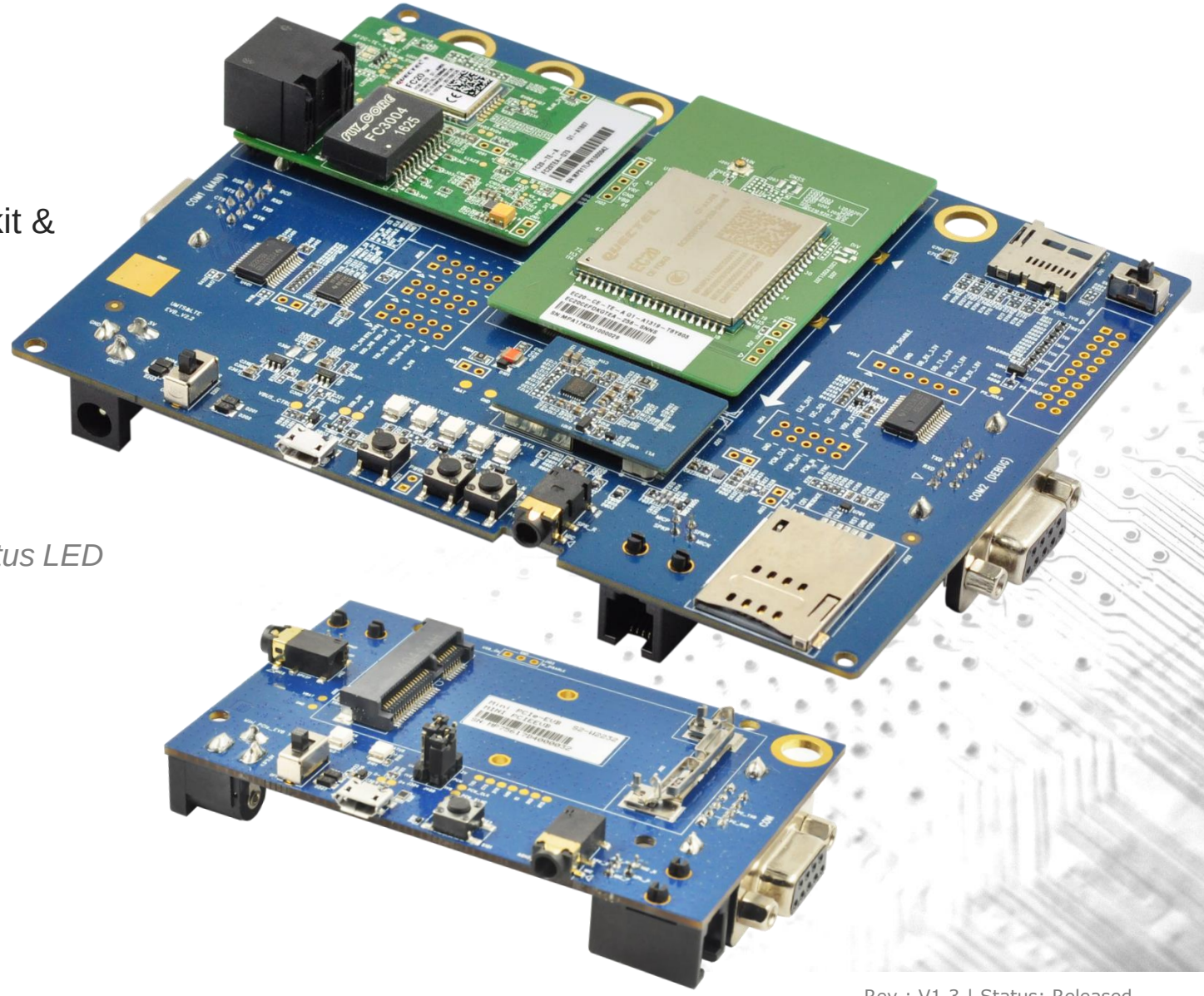
Support Package - Technical Materials & EVB Kit

Technical Materials Package

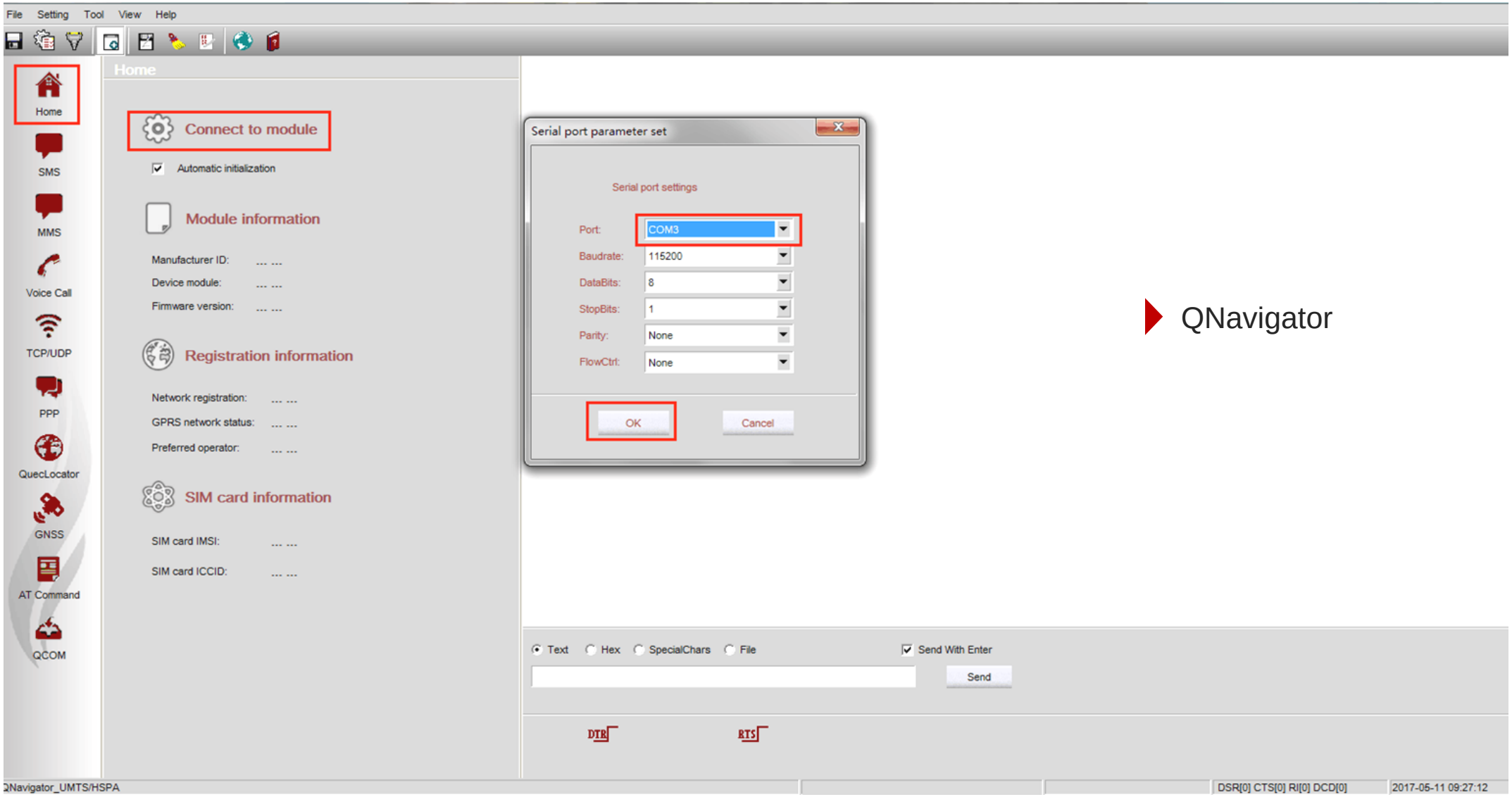
- Hardware & software documents
- Debug tools, download tools, test tools, EVB kit & USB drivers
- Approvals & test reports
- QNavigator

Development Tool (UMTS&LTE EVB Kit)

- Connectors
  a) *Power supply connectors*
  b) *USB connector*
  c) *UART connectors*
  d) *(U)SIM card connector*
  e) *SD card connector*
  f) *Digital and audio connectors*
- Auxiliaries
  a) *Network status LED*
  b) *Power key*
  c) *Reset key*
  d) *Test points*



Support Package - QNavigator



- QNavigator

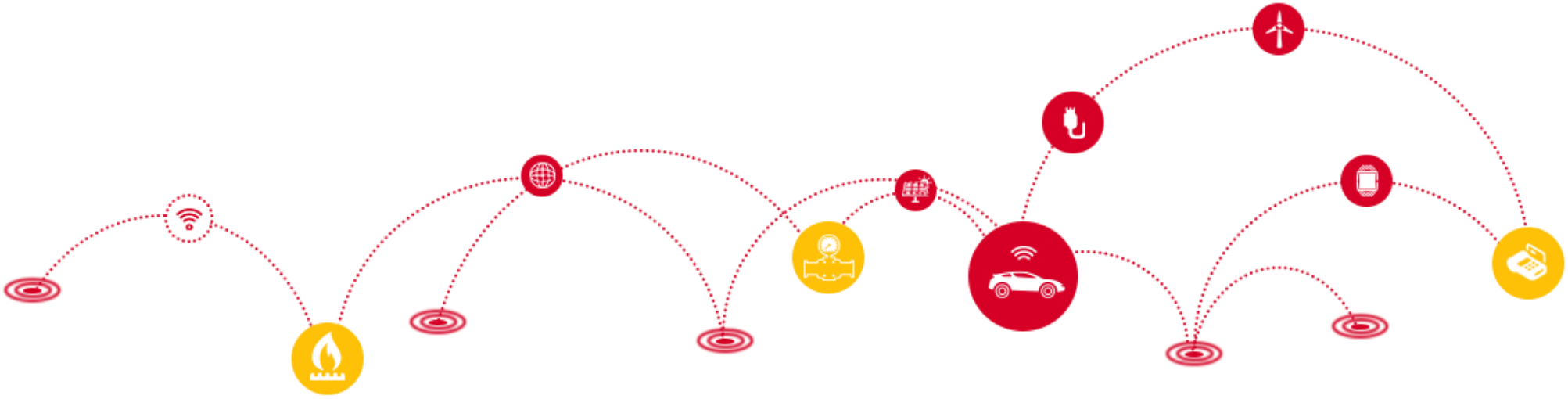
Roadmap

Specifications & Highlights

Development Timeline

Technical Details

Applications



Target Applications



Telematics & Transport

- Automotive OEM
- Vehicle Tracking
- Container Tracking
- Ship Tracking
- Fleet Management
- OBD
- DVR
- Insurance Box



Energy

- Electricity Meter
- Gas Meter
- Water Meter
- Heat Meter
- Smart Grid
- Wind Turbines
- Solar Panel
- Charging Pile



Payment

- Wireless POS
- Cash Register
- ATM
- Vending Machine
- Top-up Machine



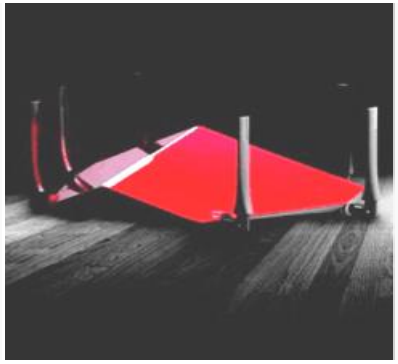
Security

- Alarm
- Camera
- Video Surveillance
- Intrusion Detection
- Smoke Detector
- Asset Protection



Smart City

- Street Lighting
- Sharing Economy
- Traffic Light
- Elevator
- Digital Signage
- Advertisement Display
- LED Lighting
- Garbage Bin Monitoring
- Parking



Gateway

- DTU
- Consumer Router
- Industrial Router
- VoIP
- Server
- Wi-Fi Hotspot



Industry

- Industrial PDA
- Rugged Tablet PC
- Pipeline Management
- UAV
- Robot
- Flow Meter
- Refrigerator
- Industrial Control
- Industrial Monitoring



Life & Healthcare

- Personal Tracker
- Household Appliances
- Pet Tracker
- Wearables
- Elderly Monitoring
- Remote Medical Equipment
- Glucometer
- Blood Pressure Monitor
- Gambling Machine



Agriculture & Environment

- Trail Camera
- Food Traceability
- Farm Machinery
- Irrigation
- Meteorological Station
- Wildlife Tracking
- Environment Monitoring
- Pollution Monitoring



Thank you!

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